

Final Examination

Fundamentals of Computing and Programming

BMath First Year

Indian Statistical Institute, Bangalore Centre

Date: 7th November, 2025, Friday | Time: 2-5 pm

Total Marks: 50

Problem 1 (5 marks)

What is the output of the following program? Explain your answer in a few lines.

```
#include<stdio.h>

#define N 100

int main(void)
{
    int i, j, a[N];
    for(a[1] = 0, i = 2; i <= N; i++)
        a[i] = 1;
    for(i = 2; i <= N / 2; i++)
        for(j = 2; j <= N / i; j++)
            a[i * j] = 0;
    for(i = 1; i <= N; i++)
        if (a[i]) printf("%d ", i);
    printf("\n");
    return 0;
}
```

Problem 2 (5 marks)

What is the output of the following program? Explain your answer in a few lines.

```
#include<stdio.h>

unsigned f(unsigned x, int p, int n)
{
    return x ^ (~(~0 << n) << (p + 1 - n));
}

int main(void)
{
    printf("%d", f(2047, 5, 3));
    return 0;
}
```

Problem 3 (5 marks)

What is the output of the following program? Explain your answer in a few lines.

```
#include<stdio.h>

int main(void)
{
    int a = 0;
    int b = 1;
    int tmp;

    while (b < 200)
    {
        tmp = b;
        b = a + b;
        a = tmp;
        printf("(%d ", a);
    }
}
```

Problem 4 (5 marks)

What is the output of the following program? Explain your answer in a few lines.

```
#include<stdio.h>
#include<string.h>

#define F(c) ('a' <= (c) && (c) <= 'z' ? (c) - 'a' + 'A' : (c))

int main(void)
{
    int i = 0;
    char *s;
    strcpy(s, "abcd");
    putchar(F(s[++i]));
    return 0;
}
```

Problem 5 (5 marks)

What is the output of the following program? Explain your answer in a few lines.

```
#include<stdio.h>

int f1(int (*f)(int));
int f2(int i);

int main(void)
{
    printf("%d\n", f1(f2));
    return 0;
}

int f1(int (*f)(int))
{
    int n = 0;
    while ((*f)(n)) n++;
    return n;
}

int f2(int n)
{
    return n*n - 5*n - 14;
}
```

Problem 6 (5 marks)

Explain the following program in a few lines. What does it do?

```
#include<stdio.h>

int main(int argc, char *argv[])
{
    int i;
    for(i = argc - 1; i > 0; i--)
        printf("%s ", argv[i]);
    printf("\n");
    return 0;
}
```

Problem 7 (6 marks)

Write the following function:

```
int *find_largest (int a[], int n);
```

When passed an array of length `n`, the function will return a pointer to the array's largest element.

Problem 8 (7 marks)

Write a function named `duplicate` that uses dynamic storage allocation to create a copy of a string. For example, the call

```
p = duplicate(str);
```

would allocate space for a string of the same length as `str`, copy the contents of `str` into the new string (using the `strcpy` function) and return a pointer to it. Have the `duplicate` function return a null pointer if the memory allocation fails.

Problem 9 (3+4 = 7 marks)

Consider the following function `f`. What is the value of `f("zxywz", "xyxz")`? In general, what value does `f` return when passed two strings `s` and `t`?

```
int f(char *s, char *t)
{
    char *p, *q;
    for(p = s; *p; p++)
    {
        for (q = t; *q; q++)
            if (*p == *q) break;
        if (*q == '\0') break;
    }
    return p - s;
}
```